

6.2 药物在体内的分布与排除





- 药物进入机体形成血药浓度 (Plasma Concentration)
- 血药浓度需保持在一定范围内——给药方案设计
- 药物动力学 (Pharmacokinetics) 定量描述药物在体内的 ADME 过程, 包括吸收 (**A**bsorption)、分布 (**D**istribution)、代谢 (**M**etabolism) 和排除 (**E**limination)

为了揭示药物的 ADME 过程的定量规律, 通常的办法是在给药后的一系列时间 t 采集血液, 并测定血药浓度 c , 通过数学方法分析 $t - c$ 数据, 还可以通过唾液中药浓度、尿药量及机体对药物的反应等进行。通过对实验数据的处理, 获取表征药物动力学特性的有关参数。

表1: 血液循环状况对药物分布的影响

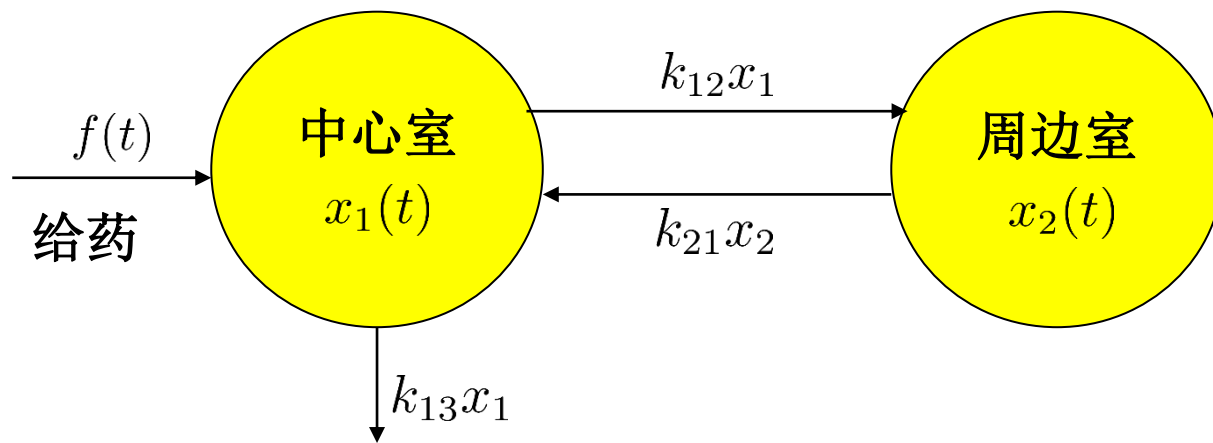
部位	肾	肝静脉	肝门静脉	心	脑	皮	脂肪
血流量 (ml/100g.min)	450	20	75	70	55	5	1

- 建立房室模型——药物动力学的基本步骤
- 房室——机体的一部分，药物在一个房室内分布均匀（血药浓度为常数），在房室间按一定规律转移
- 二室模型——中心室（心、肺、肾等）和周边室（四肢、肌肉等）

模型假设

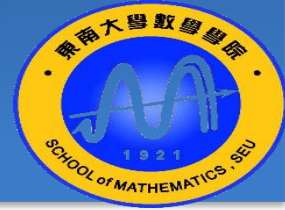
- 中心室 (1) 和周边室 (2) 容积不随时间变化
- 药物从体外先进入中心室，在二室间相互转移，最后从中心室排出体外
- ADME 过程依一级速率变化，即单位时间内转移的药物剂量与源室血浆药物剂量成正比

模型建立



$$\begin{cases} \dot{x}_1(t) = -k_{12}x_1 - k_{13}x_1 + k_{21}x_2 + f(t) \\ \dot{x}_2(t) = k_{12}x_1 - k_{21}x_2 \end{cases}$$

给药方案决定给药速率 $f(t)$ 和初始条件



模型建立

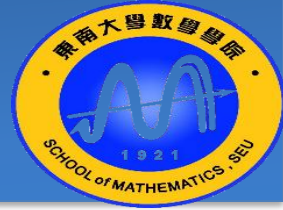
$$c_i(t) = x_i(t)/V_i, \quad i = 1, 2$$

$$\dot{c}(t) = Ac(t) + b(t) \quad (1)$$

$$A = \begin{pmatrix} -(k_{12} + k_{13}) & \frac{V_2}{V_1} k_{21} \\ \frac{V_1}{V_2} k_{12} & -k_{21} \end{pmatrix}, b(t) = \begin{pmatrix} \frac{f(t)}{V_1} \\ 0 \end{pmatrix}$$

$$\dot{c}(t) = Ac(t) \quad (2)$$

线性非齐次方程 (1) 通解 = 齐次方程通解 (2) + 非齐次方程 (1) 特解



齐次方程通解 (假设特征值互不相同)

$$\dot{c}(t) = Ac(t)$$

$$\text{通解 } c(t) = \sum_i a_i p_i e^{\lambda_i t}, a_i \in R$$

$$A = \begin{pmatrix} -(k_{12} + k_{13}) & \frac{V_2}{V_1} k_{21} \\ \frac{V_1}{V_2} k_{12} & -k_{21} \end{pmatrix}$$

$$f(\lambda) = \lambda^2 + (k_{12} + k_{13} + k_{21})\lambda + k_{21}k_{13} = 0$$

$$\lambda_1 = -\alpha, \lambda_2 = -\beta, 0 < \alpha < \beta$$

$$\text{特征向量 } p_i = ((k_{21} + \lambda_i)V_2, k_{12}V_1)^T$$

1. 快速静脉注射

$t = 0$ 瞬时注射剂量 d 的药物进入中心室, 血药浓度立即变为 d/V_1



$$\begin{cases} \dot{c}_1(t) = -(k_{12} + k_{13})c_1 + \frac{V_2}{V_1}k_{21}c_2 \\ \dot{c}_2(t) = \frac{V_1}{V_2}k_{12}c_1 - k_{21}c_2 \end{cases}, t > 0$$

$$f(t) = 0, c_1(0) = \frac{d}{V_1}, c_2(0) = 0$$

$$\begin{cases} c_1(t) = A_1 e^{-\alpha t} + B_1 e^{-\beta t} \\ c_2(t) = A_2 (e^{-\alpha t} - e^{-\beta t}) \end{cases}$$

$$A_1 = \frac{d(k_{21} - \alpha)}{V_1(\beta - \alpha)}, B_1 = \frac{d(\beta - k_{21})}{V_1(\beta - \alpha)}, A_2 = \frac{dk_{12}}{V_2(\beta - \alpha)}$$

$$\lim_{t \rightarrow \infty} c(t) = 0$$

2. 恒速静脉滴注

当 $0 \leq t \leq T$ 时, 药物以恒定速率 k 进入中心室



$$\begin{cases} \dot{c}_1(t) = -(k_{12} + k_{13})c_1 + \frac{V_2}{V_1}k_{21}c_2 + \frac{f(t)}{V_1} \\ \dot{c}_2(t) = \frac{V_1}{V_2}k_{12}c_1 - k_{21}c_2 \end{cases}, t \in [0, T]$$

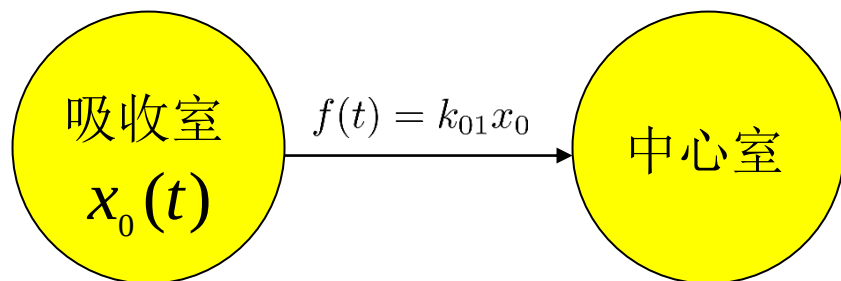
$$f(t) = k, c_1(0) = c_2(0) = 0$$

$$\begin{cases} c_1(t) = A_1 e^{-\alpha t} + B_1 e^{-\beta t} + \frac{k}{k_{13}V_1} \\ c_2(t) = A_2 e^{-\alpha t} + B_2 e^{-\beta t} + \frac{k_{12}k}{k_{21}k_{13}V_2}, 0 \leq t \leq T \\ A_1 = \frac{k(k_{13} - \beta)}{k_{13}V_1(\beta - \alpha)}, B_1 = \frac{k(\alpha - k_{13})}{k_{13}V_1(\beta - \alpha)} \\ A_2 = \frac{V_1(k_{12} + k_{13} - \alpha)}{k_{21}V_2}A_1, B_2 = \frac{V_1(k_{12} + k_{13} - \beta)}{k_{21}V_2}B_1 \end{cases}$$

$t > T$ 时, $c_1(t)$ 和 $c_2(t)$ 按指数规律衰减趋于零

3. 口服或肌肉注射

相当于药物 (剂量 d) 先进入吸收室, 吸收后再进入中心室



$$\begin{cases} \dot{x}_0(t) = -k_{01}x_0 \\ x_0(0) = d \end{cases}$$

$$x_0(t) = de^{-k_{01}t}$$

$$\begin{cases} \dot{c}_1(t) = -(k_{12} + k_{13})c_1 + \frac{V_2}{V_1}k_{21}c_2 + \frac{dk_{01}e^{-k_{01}t}}{V_1} \\ \dot{c}_2(t) = \frac{V_1}{V_2}k_{12}c_1 - k_{21}c_2 \end{cases}$$

$$c_1(t) = Ae^{-\alpha t} + Be^{-\beta t} + Ce^{-k_{01}t}$$

$$c_2(t) = \frac{k_{12}V_1}{V_2} \left[\frac{A(e^{-\alpha t} - e^{-k_{21}t})}{k_{21} - \alpha} + \frac{B(e^{-\beta t} - e^{-k_{21}t})}{k_{21} - \beta} + \frac{C(e^{-k_{01}t} - e^{-k_{21}t})}{k_{21} - k_{01}} \right]$$

$$\lim_{t \rightarrow \infty} c_1(t) = 0$$

房室模型通过对 $c_1(t)$ 的测量, 确定对药理学及临床医学最为重要的模型参数, 特别是 k_{13}

1. 参数估计

$$k_{12}, k_{21}, k_{13}, V_1, V_2$$

2. 开发应用



以快速静脉注射为例，注射剂量为 $d = 40mg$ ，在 t_i 时测得 $c_1(t_i), i = 1, 2, \sim n$.

表 1: 静脉血药浓度采集数据

时间 (h)	0.05	0.08	0.17	0.5	1	3	6	12	24
浓度 ($\mu g/ml$)	2053.8	1712.2	1064.7	200.9	49.9	34.5	27.1	16.7	6.4

方法一. 非线性最小二乘法拟合参数

$$c_1(t) = \frac{d}{V_1(\beta - \alpha)} [(k_{21} - \alpha)e^{-\alpha t} + (\beta - k_{21})e^{-\beta t}]$$

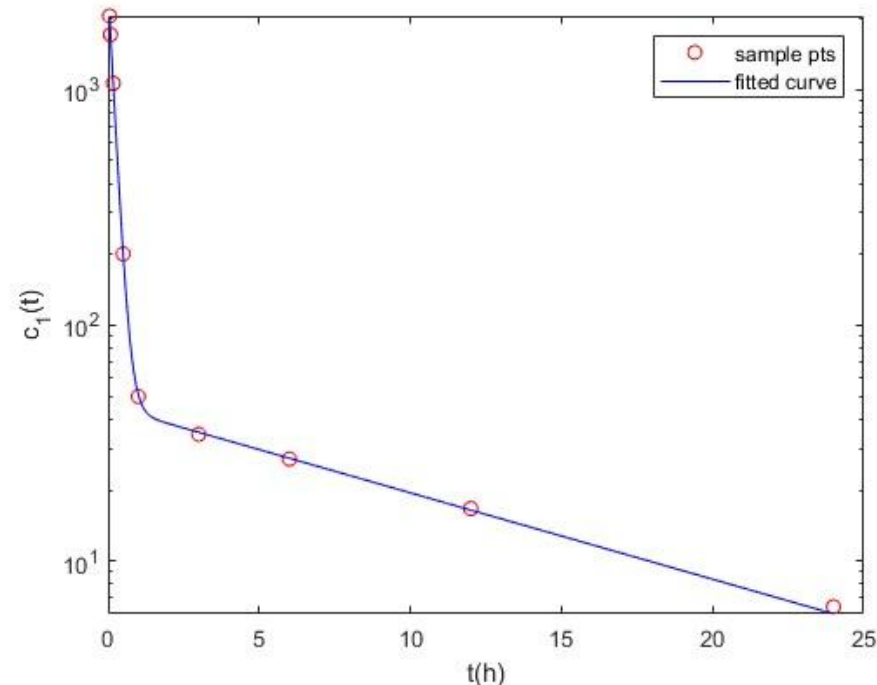
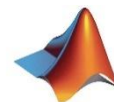
1. MATLAB 非线性拟合模型参数, $d = 40mg$

```
clear;
t=[0.05,0.08,0.17,0.5,1,3,6,12,24];
c=[2053.8,1712.2,1064.7,200.9,49.9,34.5,27.1,16.7,6.4];

[a,S]=lsqcurvefit(@c1,[7;8;9;10],t,c);

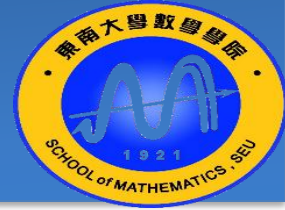
tt=min(t):0.01:max(t);
cc=c1(a,tt);
semilogy(t,c,'ro',tt,cc,'b-');
legend('sample pts','fitted curve')
xlabel('t(h)');
ylabel('c_1(t)');
```

$$\begin{aligned}\alpha &= 0.0844 \\ \beta &= 5.6570 \\ k_{21} &= 0.1781 \\ V_1 &= 14.8412\end{aligned}$$





参数估计



方法二. 线性最小二乘法分步拟合参数

$$c_1(t) = \frac{d}{V_1(\beta - \alpha)} [(k_{21} - \alpha)e^{-\alpha t} + (\beta - k_{21})e^{-\beta t}] = Ae^{-\alpha t} + Be^{-\beta t}$$

不妨设 $\alpha < \beta$, 当 t 比较大时, $c_1(t) \approx Ae^{-\alpha t}$.

Step 1. 由较大的 $t_i, c_1(t_i)$ 数据, 用线性最小二乘法拟合参数 A, α .

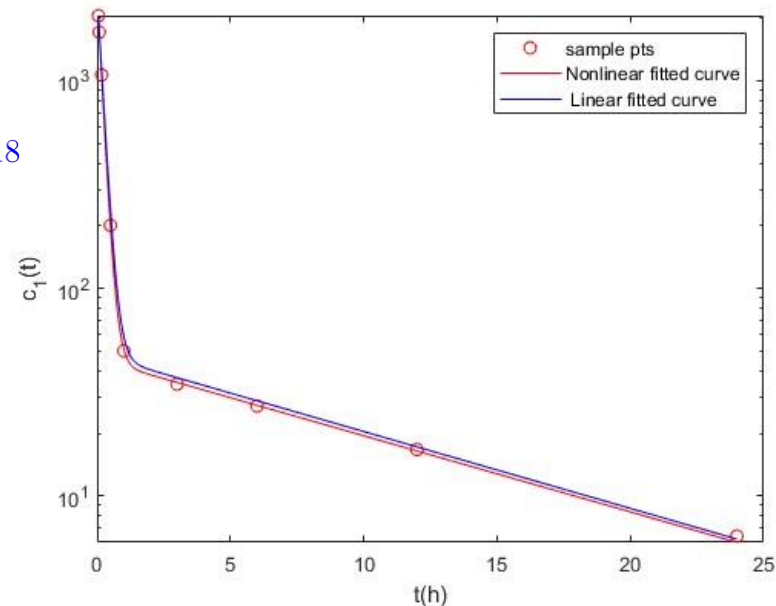
Step 2. 由较小的 $t_i, \tilde{c}_1(t_i)$ 数据, 用线性最小二乘法拟合参数 B, β . 其中

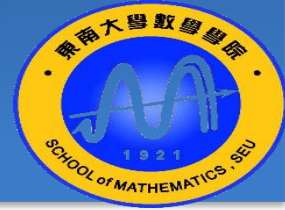
1. MATLAB 线性最小二乘拟合模型参数, $d = 40mg$

```
clear;  
t=[0.05,0.08,0.17,0.5,1,3,6,12,24];  
c=[2053.8,1712.2,1064.7,200.9,49.9,34.5,27.1,16.7,6.4];  
t1=t(5:end);  
y1=c(5:end);  
  
a=polyfit(t1,log(y1),1);  
alpha=-a(1);  
A=exp(a(2));  
  
t2=t(1:4);  
y2=c(1:4)-polyval(a,t2);  
b=polyfit(t2,log(y2),1);  
beta=-b(1);  
B=exp(b(2));  
  
figure(2)  
tt=min(t):0.01:max(t);  
cc2=A*exp(-alpha*tt)+B*exp(-beta*tt);  
semilogy(t,c,'ro',tt,cc1,'r-',tt,cc2,'b-');  
legend('sample pts','Nonlinear fitted curve','Linear fitted curve')  
xlabel('t(h)');  
ylabel('c_1(t)');
```

$$\tilde{c}_1(t) = c_1(t) - Ae^{-\alpha t} = Be^{-\beta t}$$

$$A = 47.8752, \alpha = 0.0852, B = 2.6024 \times 10^3, \beta = 5.1718$$





确定其余参数

$$\begin{cases} c_1(0) = \frac{d}{V_1} = A + B \\ d = k_{13} V_1 \int_0^{+\infty} c_1(t) dt \end{cases}$$

 $\Rightarrow$

$$\begin{aligned} V_1 &= \frac{d}{A+B} \\ k_{13} &= \frac{\alpha\beta(A+B)}{\alpha B + \beta A} \end{aligned}$$

$$\begin{cases} \alpha + \beta = k_{12} + k_{21} + k_{13} \\ \alpha\beta = k_{21}k_{13} \end{cases}$$

 $\Rightarrow$

$$\begin{aligned} k_{21} &= \frac{\alpha\beta}{k_{13}} \\ k_{12} &= \alpha + \beta - k_{13} - k_{21} \end{aligned}$$